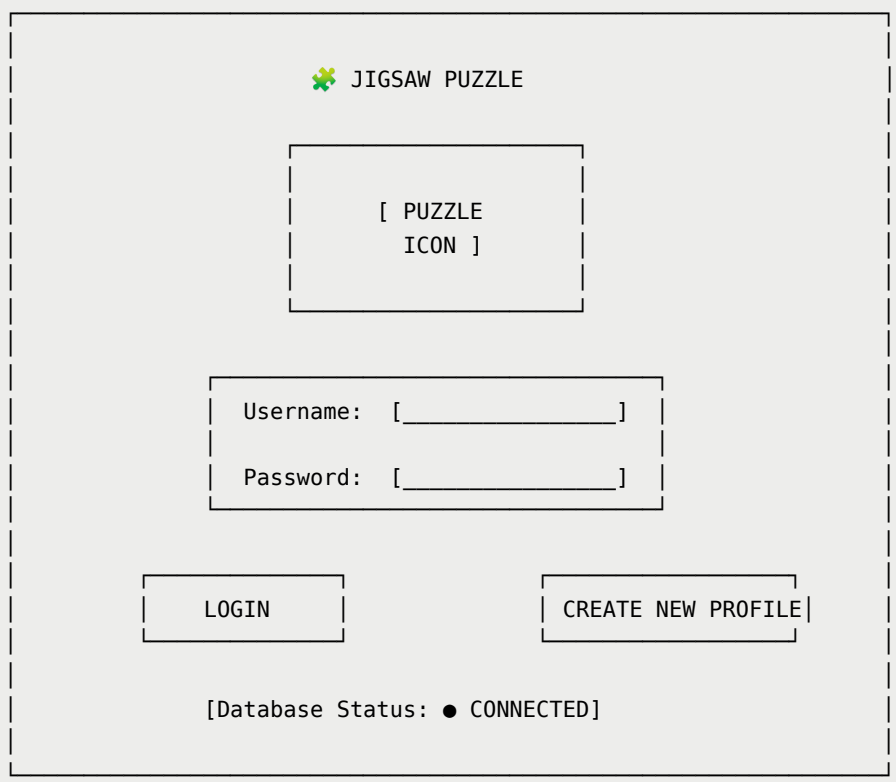
UI/UX Desgin

Page 1: Authentication Screen (App Launch)

Layout dimensions: 800×600 (centered on screen)

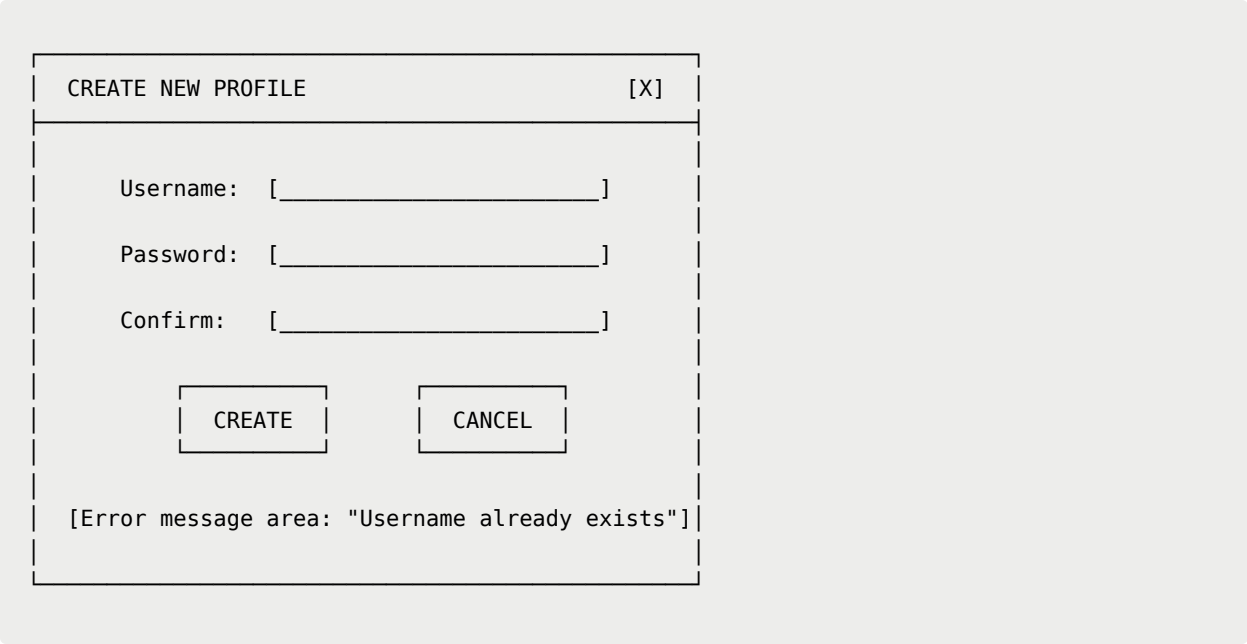


Behavior:

- App launch → attempt DB connection → show status
- Login → validates → navigates to Main Menu
- Create New Profile → opens modal dialog

Page 1a: Profile Creation Modal Dialog

Overlay on Authentication Screen

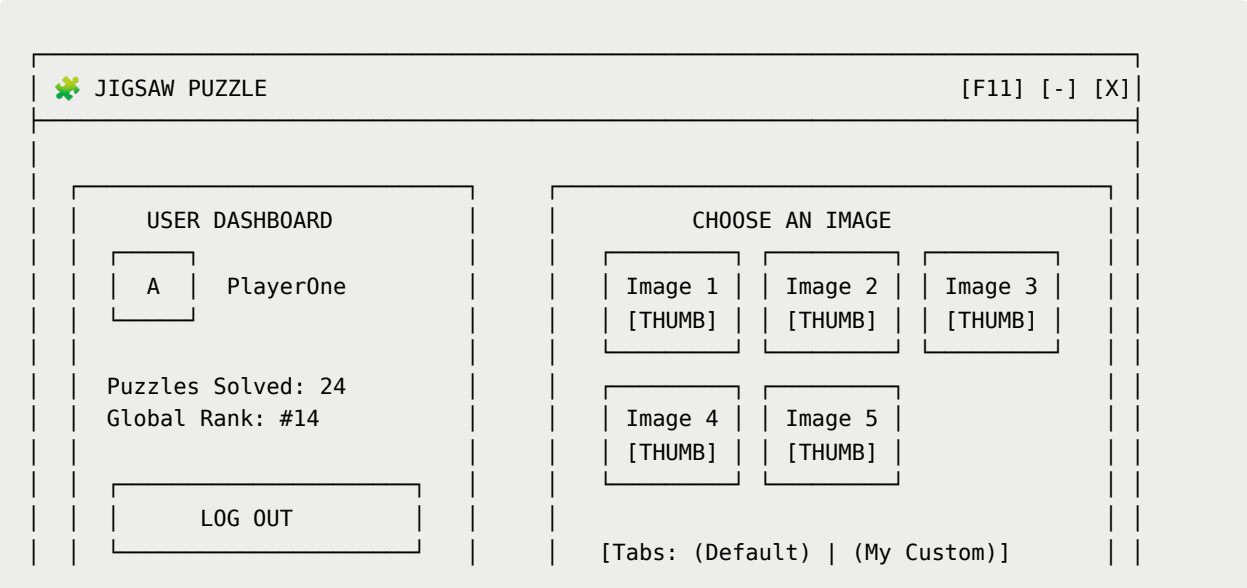


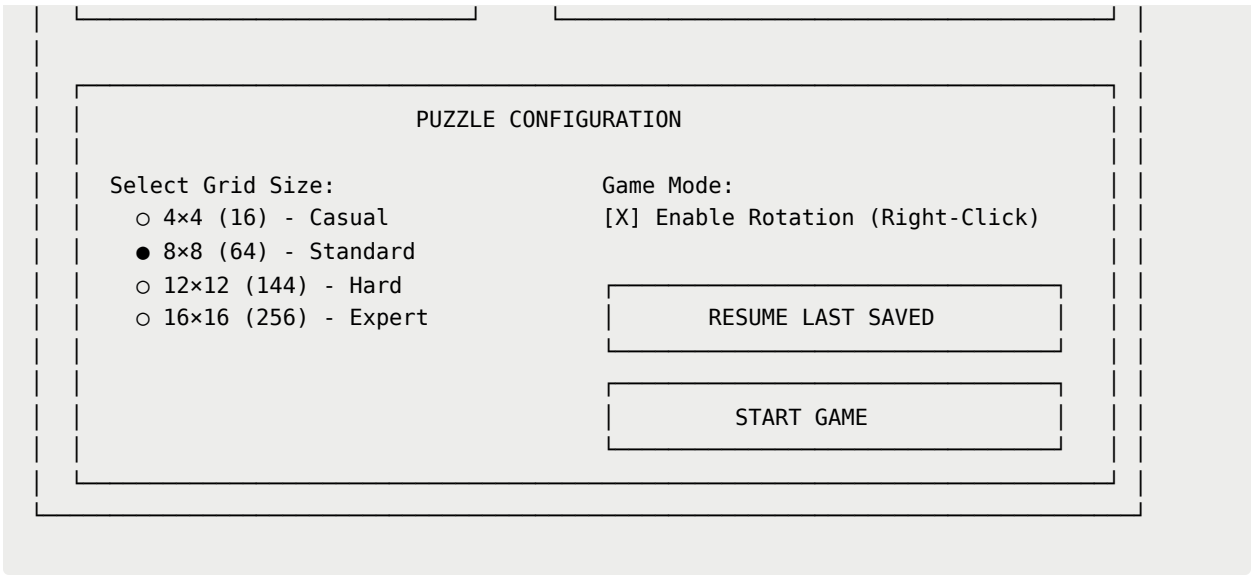
Validation:

- Username unique, non-empty
- Password non-empty, match confirm
- On success → auto-login → Main Menu

Page 2: Main Menu & Dashboard

Layout dimensions: 1280×720 (16:9)





Behavior:

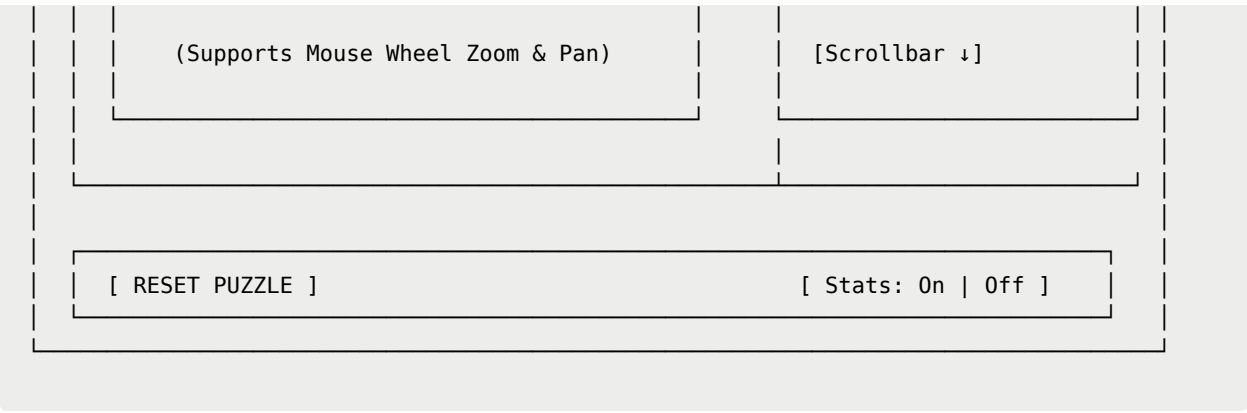
- Resume button enabled only if saved game exists for current user & selected image
- Log Out → confirmation dialog → back to Authentication Screen

---

Page 3: Main Gameplay Workspace (16:9 Optimized)

Layout dimensions: 1280×720 (fullscreen or resizable)





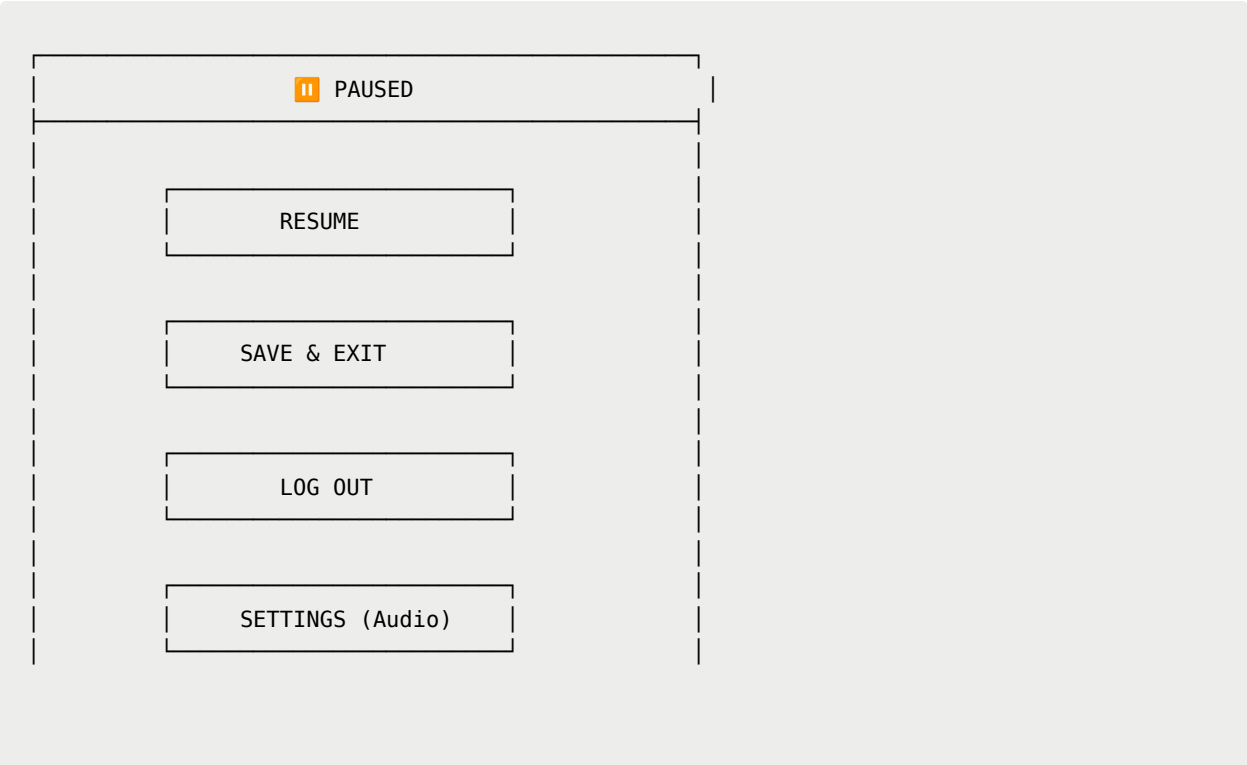
Piece Tray details:

- Scrollable vertical or horizontal panel (configurable)
- Each piece shows thumbnail preview
- Rotated pieces show visual indicator

Page 3a: Pause Menu Overlay

Trigger: Press `ESC` during gameplay

Overlay on top of Gameplay Workspace (semi-transparent dark background)



```
│                                                    │
└────────────────────────────────────────────────────┘
```

Behavior:

- Save & Exit → auto-save current state → logout → Authentication Screen
- Log Out → confirmation "Discard progress?" → no save → Authentication Screen
- Settings → volume slider, mute toggle

Page 3b: Reset Confirmation Dialog

Trigger: Click "RESET PUZZLE" button

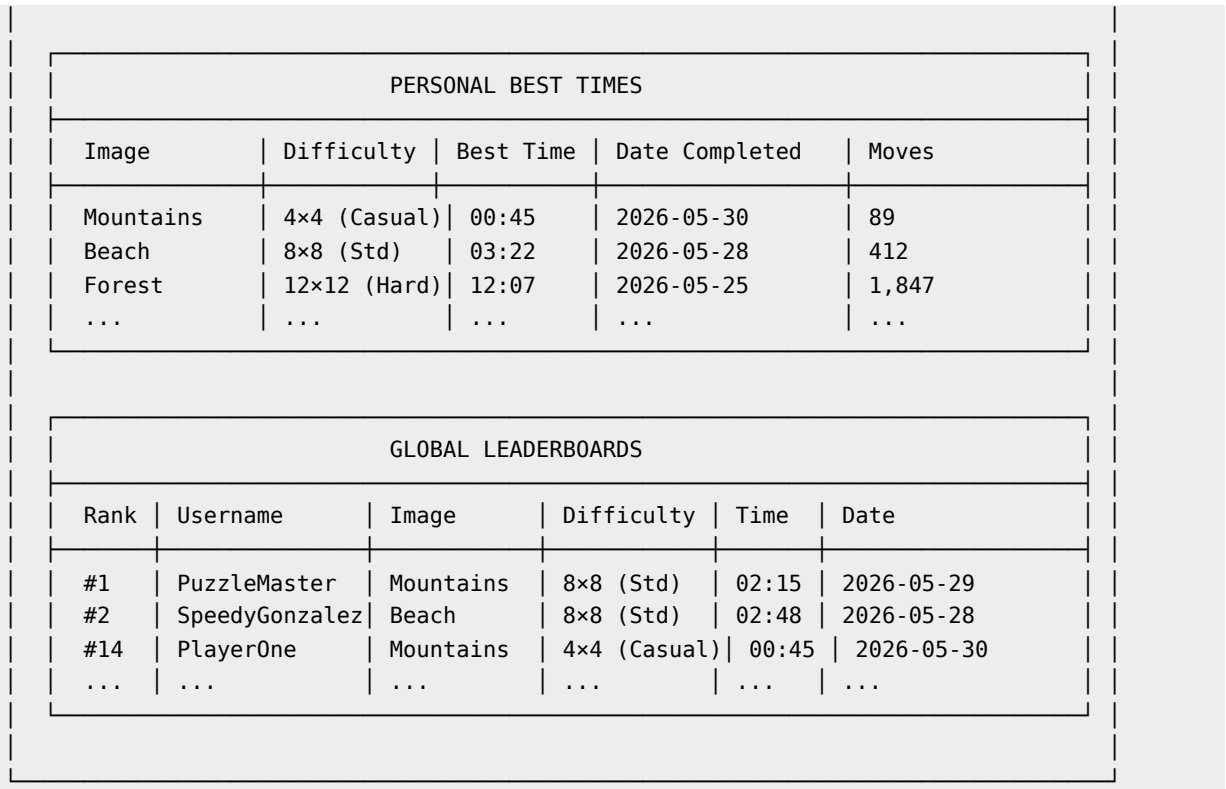
Overlay on Gameplay Workspace

```
┌────────────────────────────────────────────────────┐
│          RESET PUZZLE?                    [X]   │
├────────────────────────────────────────────────────┤
│                                                    │
│  All current progress will be lost.               │
│  Timer and move counter will reset to zero.       │
│  Previously recorded stats will remain intact.    │
│                                                    │
│         ┌──────────┐      ┌──────────┐            │
│         │  RESET   │      │  CANCEL  │            │
│         └──────────┘      └──────────┘            │
│                                                    │
└────────────────────────────────────────────────────┘
```

Page 4: Leaderboards / History Screen

Layout dimensions: 1024×768

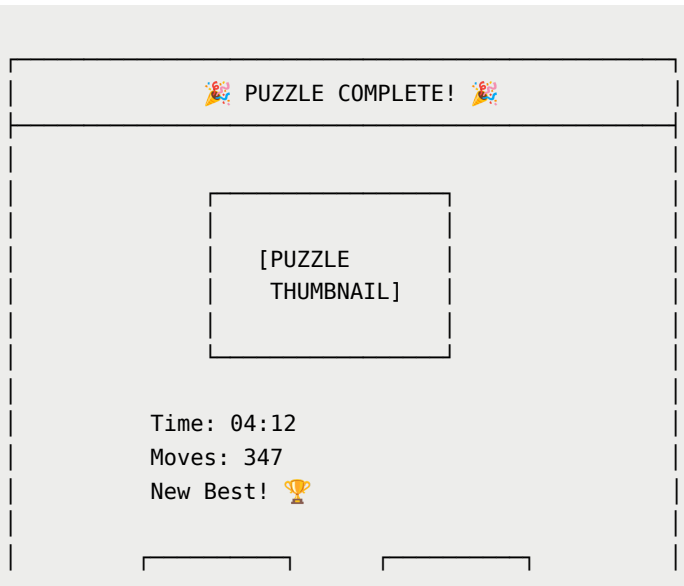
```
┌──────────────────────────────────────────────────────────────────────────────┐
│  🏆 LEADERBOARDS & HISTORY                                  [Back to Menu]  │
├──────────────────────────────────────────────────────────────────────────────┤
│                                                                              │
│  ┌──────────────────────────────────────────────────────────────────────┐   │
│  │  Filter: Image [▼ All Images]          Difficulty [▼ All Difficulties]  │   │
│  └──────────────────────────────────────────────────────────────────────┘   │
```

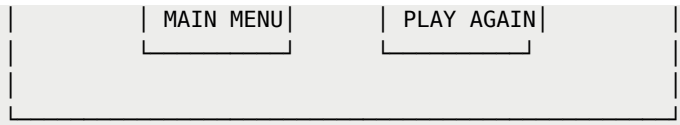


Page 5: Puzzle Completion Dialog

Trigger: Last piece snapped into correct position

Overlay on Gameplay Workspace





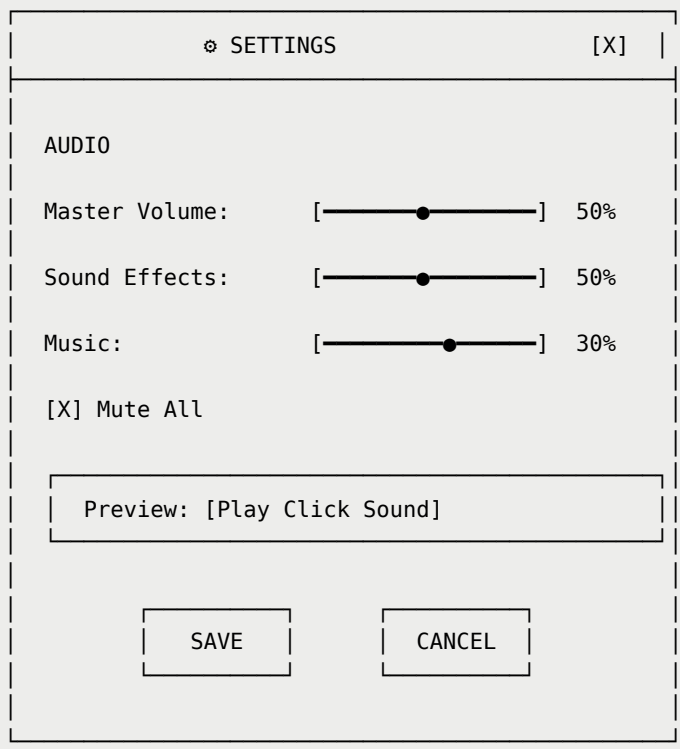
Behavior:

- Updates PlayerStats table (increment PuzzlesSolved, update BestTime if faster)
- Play Again → returns to Main Menu with same image/difficulty preselected
- Main Menu → returns to dashboard

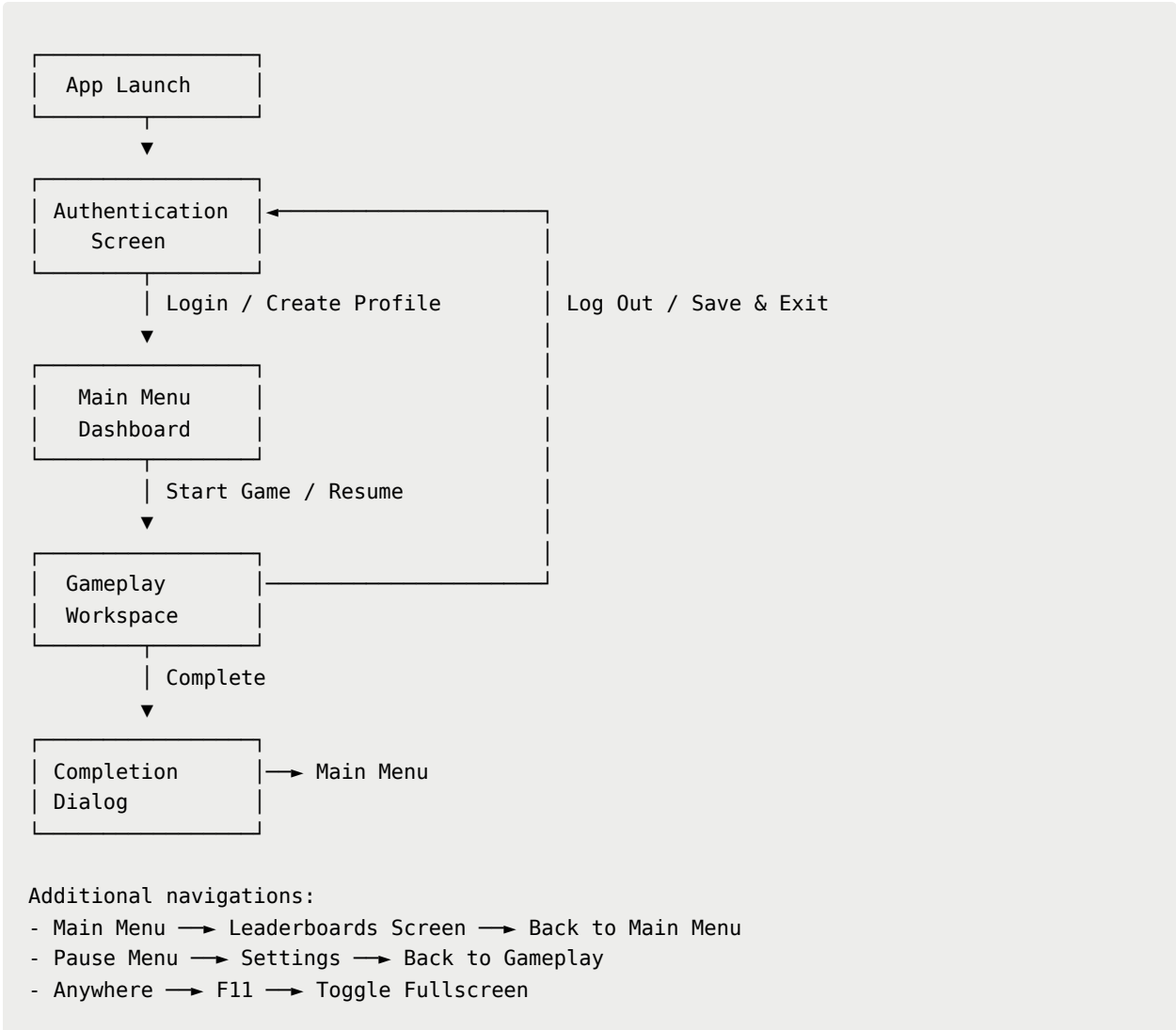
Page 6: Settings / Audio Panel

Access: From Pause Menu → Settings

Overlay on current screen



Navigation Flow Diagram



Summary: All Pages

Page #	Screen Name	Purpose	Navigation From	Navigation To
1	Authentication	Login / Profile creation	App launch	Main Menu
1a	Profile Modal	Create new user	Auth Screen (Create button)	Main Menu (on success)
2	Main Menu	Dashboard, image selection,	Auth Screen, Completion	Gameplay, Leaderboards,

Page #	Screen Name	Purpose	Navigation From	Navigation To
		difficulty	Dialog	Auth (logout)
3	Gameplay Workspace	Main puzzle gameplay	Main Menu	Completion Dialog, Auth (logout)
3a	Pause Menu	Resume, save & exit, logout, settings	Gameplay (ESC)	Gameplay, Auth, Settings
3b	Reset Confirmation	Confirm puzzle reset	Gameplay (Reset button)	Gameplay (reset)
4	Leaderboards	Personal & global best times	Main Menu	Main Menu
5	Completion Dialog	Show stats after puzzle solved	Gameplay (completion)	Main Menu
6	Settings	Audio configuration	Pause Menu	Pause Menu

This completes the **end-to-end UI layout specification** for all pages required by your SRS.